Troubleshooting lost and missing object data

StorageGRID 11.5

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Troubleshooting lost and missing object data

Objects can be retrieved for several reasons, including read requests from a client application, background verifications of replicated object data, ILM re-evaluations, and the restoration of object data during the recovery of a Storage Node.

The StorageGRID system uses location information in an object's metadata to determine from which location to retrieve the object. If a copy of the object is not found in the expected location, the system attempts to retrieve another copy of the object from elsewhere in the system, assuming that the ILM policy contains a rule to make two or more copies of the object.

If this retrieval is successful, the StorageGRID system replaces the missing copy of the object. Otherwise, the **Objects lost** alert and the legacy LOST (Lost Objects) alarm are triggered, as follows:

- For replicated copies, if another copy cannot be retrieved, the object is considered lost, and the alert and alarm are triggered.
- For erasure coded copies, if a copy cannot be retrieved from the expected location, the Corrupt Copies Detected (ECOR) attribute is incremented by one before an attempt is made to retrieve a copy from another location. If no other copy is found, the alert and alarm are triggered.

You should investigate all **Objects lost** alerts immediately to determine the root cause of the loss and to determine if the object might still exist in an offline, or otherwise currently unavailable, Storage Node or Archive Node.

In the case where object data without copies is lost, there is no recovery solution. However, you must reset the Lost Object counter to prevent known lost objects from masking any new lost objects.

Related information

Investigating lost objects

Resetting lost and missing object counts

Investigating lost objects

When the **Objects lost** alert and the legacy LOST (Lost Objects) alarm are triggered, you must investigate immediately. Collect information about the affected objects and contact technical support.

What you'll need

- You must be signed in to the Grid Manager using a supported browser.
- You must have specific access permissions.
- You must have the `Passwords.txt` file.

About this task

The **Objects lost** alert and the LOST alarm indicate that StorageGRID believes that there are no copies of an object in the grid. Data might have been permanently lost.

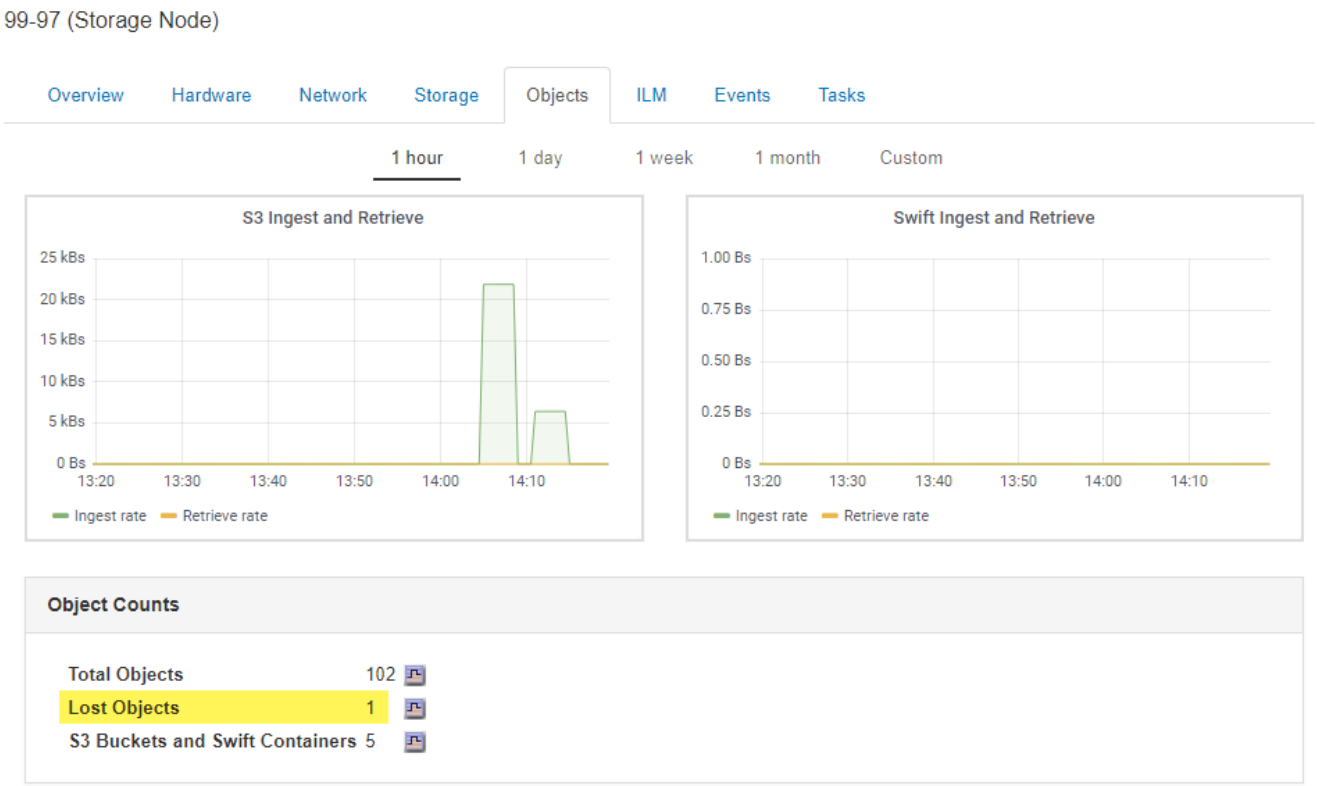
Investigate lost object alarms or alerts immediately. You might need to take action to prevent further data loss. In some cases, you might be able to restore a lost object if you take prompt action.

The number of Lost Objects can be seen in the Grid Manager.

Steps

1. Select **Nodes**.
2. Select ***Storage Node*** > **Objects**.
3. Review the number of Lost Objects shown in the Object Counts table.

   This number indicates the total number of objects this grid node detects as missing from the entire StorageGRID system. The value is the sum of the Lost Objects counters of the Data Store component within the LDR and DDS services.



4. From an Admin Node, access the audit log to determine the unique identifier (UUID) of the object that triggered the **Objects lost** alert and the LOST alarm:
   a. Log in to the grid node:
      i. Enter the following command: `ssh admin@grid_node_IP`
      ii. Enter the password listed in the `Passwords.txt` file.
      iii. Enter the following command to switch to root: `su -`
      iv. Enter the password listed in the `Passwords.txt` file. When you are logged in as root, the prompt changes from `$` to `#`.
   b. Change to the directory where the audit logs are located. Enter: `cd /var/local/audit/export/`
   c. Use grep to extract the Object Lost (OLST) audit messages. Enter: `grep OLST audit_file_name`
   d. Note the UUID value included in the message.

```
>Admin: # grep OLST audit.log
2020-02-12T19:18:54.780426
[AUDT:[CBID(UI64):0x38186FE53E3C49A5][UUID(CSTR):926026C4-00A4-449B-
AC72-BCCA72DD1311]
[PATH(CSTR):"source/cats"][NOID(UI32):12288733][VOLI(UI64):3222345986
][RSLT(FC32):NONE][AVER(UI32):10]
[ATIM(UI64):1581535134780426][ATYP(FC32):OLST][ANID(UI32):12448208][A
MID(FC32):ILMX][ATID(UI64):7729403978647354233]]
```

5. Use the `ObjectByUUID` command to find the object by its identifier (UUID), and then determine if data is at risk.
   a. Telnet to localhost 1402 to access the LDR console.
   b. Enter: `/proc/OBRP/ObjectByUUID UUID_value`

      In this first example, the object with `UUID 926026C4-00A4-449B-AC72-BCCA72DD1311` has two locations listed.

```
ade 12448208: /proc/OBRP > ObjectByUUID 926026C4-00A4-449B-AC72-
BCCA72DD1311

{
    "TYPE(Object Type)": "Data object",
    "CHND(Content handle)": "926026C4-00A4-449B-AC72-BCCA72DD1311",
    "NAME": "cats",
    "CBID": "0x38186FE53E3C49A5",
    "PHND(Parent handle, UUID)": "221CABD0-4D9D-11EA-89C3-
ACBB00BB82DD",
    "PPTH(Parent path)": "source",
    "META": {
        "BASE(Protocol metadata)": {
            "PAWS(S3 protocol version)": "2",
            "ACCT(S3 account ID)": "44084621669730638018",
            "*ctp(HTTP content MIME type)": "binary/octet-stream"
        },
        "BYCB(System metadata)": {
            "CSIZ(Plaintext object size)": "5242880",
            "SHSH(Supplementary Plaintext hash)": "MD5D
0xBAC2A2617C1DFF7E959A76731E6EAF5E",
            "BSIZ(Content block size)": "5252084",
            "CVER(Content block version)": "196612",
            "CTME(Object store begin timestamp)": "2020-02-
12T19:16:10.983000",
            "MTME(Object store modified timestamp)": "2020-02-
12T19:16:10.983000",
```

```
            "ITME": "1581534970983000"
        },
        "CMSM": {
            "LATM(Object last access time)": "2020-02-
12T19:16:10.983000"
        },
        "AWS3": {
            "LOCC": "us-east-1"
        }
    },
    "CLCO\(Locations\)": \[
        \{
            "Location Type": "CLDI\(Location online\)",
            "NOID\(Node ID\)": "12448208",
            "VOLI\(Volume ID\)": "3222345473",
            "Object File Path":
"/var/local/rangedb/1/p/17/11/00rH0%DkRt78Ila\#3udu",
            "LTIM\(Location timestamp\)": "2020-02-
12T19:36:17.880569"
        \},
        \{
            "Location Type": "CLDI\(Location online\)",
            "NOID\(Node ID\)": "12288733",
            "VOLI\(Volume ID\)": "3222345984",
            "Object File Path":
"/var/local/rangedb/0/p/19/11/00rH0%DkRt78Rrb\#3s;L",
            "LTIM\(Location timestamp\)": "2020-02-
12T19:36:17.934425"
        }
    ]
}
```


In the second example, the object with `UUID 926026C4-00A4-449B-AC72-BCCA72DD1311` has no locations listed.

```
ade 12448208: / > /proc/OBRP/ObjectByUUID 926026C4-00A4-449B-AC72-
BCCA72DD1311

{
    "TYPE(Object Type)": "Data object",
    "CHND(Content handle)": "926026C4-00A4-449B-AC72-BCCA72DD1311",
    "NAME": "cats",
    "CBID": "0x38186FE53E3C49A5",
    "PHND(Parent handle, UUID)": "221CABD0-4D9D-11EA-89C3-
ACBB00BB82DD",
    "PPTH(Parent path)": "source",
    "META": {
        "BASE(Protocol metadata)": {
            "PAWS(S3 protocol version)": "2",
            "ACCT(S3 account ID)": "44084621669730638018",
            "*ctp(HTTP content MIME type)": "binary/octet-stream"
        },
        "BYCB(System metadata)": {
            "CSIZ(Plaintext object size)": "5242880",
            "SHSH(Supplementary Plaintext hash)": "MD5D
0xBAC2A2617C1DFF7E959A76731E6EAF5E",
            "BSIZ(Content block size)": "5252084",
            "CVER(Content block version)": "196612",
            "CTME(Object store begin timestamp)": "2020-02-
12T19:16:10.983000",
            "MTME(Object store modified timestamp)": "2020-02-
12T19:16:10.983000",
            "ITME": "1581534970983000"
        },
        "CMSM": {
            "LATM(Object last access time)": "2020-02-
12T19:16:10.983000"
        },
        "AWS3": {
            "LOCC": "us-east-1"
        }
    }
}
```

c. Review the output of /proc/OBRP/ObjectByUUID, and take the appropriate action:

Metadata	Conclusion
No object found ("ERROR":"")	If the object is not found, the message "ERROR":"" is returned. If the object is not found, it is safe to ignore the alarm. The lack of an object indicates that the object was intentionally deleted.
Locations > 0	If there are locations listed in the output, the Lost Objects alarm might be a false positive. Confirm that the objects exist. Use the Node ID and filepath listed in the output to confirm that the object file is in the listed location. (The procedure for finding potentially lost objects explains how to use the Node ID to find the correct Storage Node.) Searching for and restoring potentially lost objects If the objects exist, you can reset the count of Lost Objects to clear the alarm and the alert.
Locations = 0	If there are no locations listed in the output, the object is potentially missing. You can try to find and restore the object yourself, or you can contact technical support. Searching for and restoring potentially lost objects Technical support might ask you to determine if there is a storage recovery procedure in progress. That is, has a *repair-data* command been issued on any Storage Node, and is the recovery still in progress? See the information about restoring object data to a storage volume in the recovery and maintenance instructions.

Related information

Maintain & recover

Review audit logs

Searching for and restoring potentially lost objects

It might be possible to find and restore objects that have triggered a Lost Objects (LOST) alarm and a **Object lost** alert and that you have identified as potentially lost.

What you'll need

- You must have the UUID of any lost object, as identified in "Investigating lost objects."
- You must have the `Passwords.txt` file.

About this task

You can follow this procedure to look for replicated copies of the lost object elsewhere in the grid. In most cases, the lost object will not be found. However, in some cases, you might be able to find and restore a lost

replicated object if you take prompt action.

> Contact technical support for assistance with this procedure.

Steps

1. From an Admin Node, search the audit logs for possible object locations:
   a. Log in to the grid node:
      i. Enter the following command: `ssh admin@grid_node_IP`
      ii. Enter the password listed in the `Passwords.txt` file.
      iii. Enter the following command to switch to root: `su -`
      iv. Enter the password listed in the `Passwords.txt` file. When you are logged in as root, the prompt changes from `$` to `#`.
   b. Change to the directory where the audit logs are located: `cd /var/local/audit/export/`
   c. Use grep to extract the audit messages associated with the potentially lost object and send them to an output file. Enter: `grep uuid-valueaudit_file_name > output_file_name`

      For example:

      ```
      Admin: # grep 926026C4-00A4-449B-AC72-BCCA72DD1311 audit.log >
      messages_about_lost_object.txt
      ```

   d. Use grep to extract the Location Lost (LLST) audit messages from this output file. Enter: `grep LLST output_file_name`

      For example:

      ```
      Admin: # grep LLST messages_about_lost_objects.txt
      ```

      An LLST audit message looks like this sample message.

      ```
      [AUDT:\[NOID\(UI32\):12448208\][CBIL(UI64):0x38186FE53E3C49A5]
      [UUID(CSTR):"926026C4-00A4-449B-AC72-BCCA72DD1311"][LTYP(FC32):CLDI]
      [PCLD\(CSTR\):"/var/local/rangedb/1/p/17/11/00rH0%DkRs&LgA%\#3tN6"\]
      [TSRC(FC32):SYST][RSLT(FC32):NONE][AVER(UI32):10][ATIM(UI64):
      1581535134379225][ATYP(FC32):LLST][ANID(UI32):12448208][AMID(FC32):CL
      SM]
      [ATID(UI64):7086871083190743409]]
      ```

   e. Find the PCLD field and the NOID field in the LLST message.

      If present, the value of PCLD is the complete path on disk to the missing replicated object copy. The value of NOID is the node id of the LDR where a copy of the object might be found.

If you find an object location, you might be able to restore the object.

f. Find the Storage Node for this LDR node ID.

There are two ways to use the node ID to find the Storage Node:

- In the Grid Manager, select **Support** > **Tools** > **Grid Topology**. Then select ***Data Center*** > ***Storage Node*** > **LDR**. The LDR node ID is in the Node Information table. Review the information for each Storage Node until you find the one that hosts this LDR.
- Download and unzip the Recovery Package for the grid. There is a *\docs* directory in the SAID package. If you open the index.html file, the Servers Summary shows all node IDs for all grid nodes.

2. Determine if the object exists on the Storage Node indicated in the audit message:

a. Log in to the grid node:

i. Enter the following command: `ssh admin@grid_node_IP`

ii. Enter the password listed in the `Passwords.txt` file.

iii. Enter the following command to switch to root: `su -`

iv. Enter the password listed in the `Passwords.txt` file.

When you are logged in as root, the prompt changes from `$` to `#`.

b. Determine if the file path for the object exists.

For the file path of the object, use the value of PCLD from the LLST audit message.

For example, enter:

```
ls '/var/local/rangedb/1/p/17/11/00rH0%DkRs&LgA%#3tN6'
```

Note: Always enclose the object file path in single quotes in commands to escape any special characters.

- If the object path is not found, the object is lost and cannot be restored using this procedure. Contact technical support.
- If the object path is found, continue with step Restore the object to StorageGRID. You can attempt to restore the found object back to StorageGRID.

3. If the object path was found, attempt to restore the object to StorageGRID:

a. From the same Storage Node, change the ownership of the object file so that it can be managed by StorageGRID. Enter: `chown ldr-user:bycast 'file_path_of_object'`

b. Telnet to localhost 1402 to access the LDR console. Enter: `telnet 0 1402`

c. Enter: `cd /proc/STOR`

d. Enter: `Object_Found 'file_path_of_object'`

For example, enter:

```
Object_Found '/var/local/rangedb/1/p/17/11/00rH0%DkRs&LgA%#3tN6'
```

Issuing the `Object\_Found` command notifies the grid of the object's location. It also triggers the active ILM policy, which makes additional copies as specified in the policy.

Note: If the Storage Node where you found the object is offline, you can copy the object to any Storage Node that is online. Place the object in any /var/local/rangedb directory of the online Storage Node. Then, issue the `Object\_Found` command using that file path to the object.

- If the object cannot be restored, the `Object\_Found` command fails. Contact technical support.
- If the object was successfully restored to StorageGRID, a success message appears. For example:

```
ade 12448208: /proc/STOR > Object_Found
'/var/local/rangedb/1/p/17/11/00rH0%DkRs&LgA%#3tN6'

ade 12448208: /proc/STOR > Object found succeeded.
First packet of file was valid. Extracted key: 38186FE53E3C49A5
Renamed '/var/local/rangedb/1/p/17/11/00rH0%DkRs&LgA%#3tN6' to
'/var/local/rangedb/1/p/17/11/00rH0%DkRt78Ila#3udu'
```

Continue with step Verify that new locations were created

4. If the object was successfully restored to StorageGRID, verify that new locations were created.
   a. Enter: `cd /proc/OBRP`
   b. Enter: `ObjectByUUID UUID_value`

   The following example shows that there are two locations for the object with UUID 926026C4-00A4-449B-AC72-BCCA72DD1311.

```
ade 12448208: /proc/OBRP > ObjectByUUID 926026C4-00A4-449B-AC72-
BCCA72DD1311

{
    "TYPE(Object Type)": "Data object",
    "CHND(Content handle)": "926026C4-00A4-449B-AC72-BCCA72DD1311",
    "NAME": "cats",
    "CBID": "0x38186FE53E3C49A5",
    "PHND(Parent handle, UUID)": "221CABD0-4D9D-11EA-89C3-
ACBB00BB82DD",
    "PPTH(Parent path)": "source",
    "META": {
        "BASE(Protocol metadata)": {
            "PAWS(S3 protocol version)": "2",
            "ACCT(S3 account ID)": "44084621669730638018",
```

```
            "*ctp(HTTP content MIME type)": "binary/octet-stream"
        },
        "BYCB(System metadata)": {
            "CSIZ(Plaintext object size)": "5242880",
            "SHSH(Supplementary Plaintext hash)": "MD5D
0xBAC2A2617C1DFF7E959A76731E6EAF5E",
            "BSIZ(Content block size)": "5252084",
            "CVER(Content block version)": "196612",
            "CTME(Object store begin timestamp)": "2020-02-
12T19:16:10.983000",
            "MTME(Object store modified timestamp)": "2020-02-
12T19:16:10.983000",
            "ITME": "1581534970983000"
        },
        "CMSM": {
            "LATM(Object last access time)": "2020-02-
12T19:16:10.983000"
        },
        "AWS3": {
            "LOCC": "us-east-1"
        }
    },
    "CLCO\(Locations\)": \[
        \{
            "Location Type": "CLDI\(Location online\)",
            "NOID\(Node ID\)": "12448208",
            "VOLI\(Volume ID\)": "3222345473",
            "Object File Path":
"/var/local/rangedb/1/p/17/11/00rH0%DkRt78Ila\#3udu",
            "LTIM\(Location timestamp\)": "2020-02-
12T19:36:17.880569"
        \},
        \{
            "Location Type": "CLDI\(Location online\)",
            "NOID\(Node ID\)": "12288733",
            "VOLI\(Volume ID\)": "3222345984",
            "Object File Path":
"/var/local/rangedb/0/p/19/11/00rH0%DkRt78Rrb\#3s;L",
            "LTIM\(Location timestamp\)": "2020-02-
12T19:36:17.934425"
        }
    ]
}
```


c. Sign out of the LDR console. Enter: `exit`

5. From an Admin Node, search the audit logs for the ORLM audit message for this object to confirm that information lifecycle management (ILM) has placed copies as required.
   a. Log in to the grid node:
      i. Enter the following command: `ssh admin@grid_node_IP`
      ii. Enter the password listed in the `Passwords.txt` file.
      iii. Enter the following command to switch to root: `su -`
      iv. Enter the password listed in the `Passwords.txt` file. When you are logged in as root, the prompt changes from `$` to `#`.
   b. Change to the directory where the audit logs are located: `cd /var/local/audit/export/`
   c. Use grep to extract the audit messages associated with the object to an output file. Enter: `grep uuid-valueaudit_file_name > output_file_name`

      For example:

      ```
      Admin: # grep 926026C4-00A4-449B-AC72-BCCA72DD1311 audit.log >
      messages_about_restored_object.txt
      ```

   d. Use grep to extract the Object Rules Met (ORLM) audit messages from this output file. Enter: `grep ORLM output_file_name`

      For example:

      ```
      Admin: # grep ORLM messages_about_restored_object.txt
      ```

      An ORLM audit message looks like this sample message.

      ```
      [AUDT:[CBID(UI64):0x38186FE53E3C49A5][RULE(CSTR):"Make 2 Copies"]
      [STAT(FC32):DONE][CSIZ(UI64):0][UUID(CSTR):"926026C4-00A4-449B-AC72-
      BCCA72DD1311"]
      [LOCS(CSTR):"**CLDI 12828634 2148730112**, CLDI 12745543 2147552014"]
      [RSLT(FC32):SUCS][AVER(UI32):10][ATYP(FC32):ORLM][ATIM(UI64):15633982
      30669]
      [ATID(UI64):15494889725796157557][ANID(UI32):13100453][AMID(FC32):BCM
      S]]
      ```

   e. Find the LOCS field in the audit message.

      If present, the value of CLDI in LOCS is the node ID and the volume ID where an object copy has been created. This message shows that the ILM has been applied and that two object copies have been created in two locations in the grid.

   f. Reset the count of lost objects in the Grid Manager.

Related information

Investigating lost objects

Confirming object data locations

Resetting lost and missing object counts

Review audit logs

Resetting lost and missing object counts

After investigating the StorageGRID system and verifying that all recorded lost objects are permanently lost or that it is a false alarm, you can reset the value of the Lost Objects attribute to zero.

What you'll need

- You must be signed in to the Grid Manager using a supported browser.
- You must have specific access permissions.

About this task

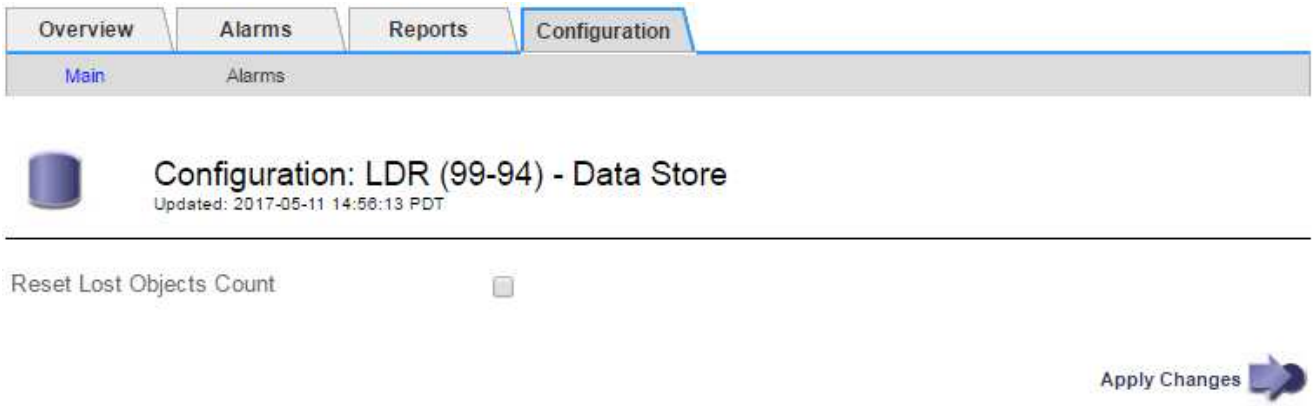
You can reset the Lost Objects counter from either of the following pages:

- **Support** > **Tools** > **Grid Topology** > ***site*** **>** ***Storage Node*** > **LDR** > **Data Store** > **Overview** > **Main**
- **Support** > **Tools** > **Grid Topology** > ***site*** **>** ***Storage Node*** > **DDS** > **Data Store** > **Overview** > **Main**

These instructions show resetting the counter from the **LDR** > **Data Store** page.

Steps

1. Select **Support** > **Tools** > **Grid Topology**.
2. Select ***Site*** **>** ***Storage Node*** > **LDR** > **Data Store** > **Configuration** for the Storage Node that has the **Objects lost** alert or the LOST alarm.
3. Select **Reset Lost Objects Count**.



4. Click **Apply Changes**.

   The Lost Objects attribute is reset to 0 and the **Objects lost** alert and the LOST alarm clear, which can take a few minutes.

5. Optionally, reset other related attribute values that might have been incremented in the process of

identifying the lost object.

a. Select ***Site* > *Storage Node*** > **LDR** > **Erasure Coding** > **Configuration**.
b. Select **Reset Reads Failure Count** and **Reset Corrupt Copies Detected Count**.
c. Click **Apply Changes**.
d. Select ***Site* > *Storage Node*** > **LDR** > **Verification** > **Configuration**.
e. Select **Reset Missing Objects Count** and **Reset Corrupt Objects Count**.
f. If you are confident that quarantined objects are not required, you can select **Delete Quarantined Objects**.

   Quarantined objects are created when background verification identifies a corrupt replicated object copy. In most cases StorageGRID automatically replaces the corrupt object, and it is safe to delete the quarantined objects. However, if the **Objects lost** alert or the LOST alarm is triggered, technical support might want to access the quarantined objects.

g. Click **Apply Changes**.

It can take a few moments for the attributes to reset after you click **Apply Changes**.

Related information

Administer StorageGRID

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